

ACADEMIC PROJECT • ANDROID APPLICATION

Arogya Mitra

Indigenous Medicine Portal

Bridging Traditional Ayurvedic Wisdom with Modern AI Technology

Developed by: Vibhu et al.

Platform: Android (Java) • AI: TensorFlow Lite • Backend: Firebase • Version: 1.0 • April 2026

Problem Statement

Why does Arogya Mitra need to exist?



Vanishing Traditional Knowledge

Indigenous medicinal plant knowledge from North-East India is passed down orally. With each generation, critical information about plant identification, preparation, and uses is permanently lost.



No Accessible Digital Repository

Practitioners and learners lack a unified, searchable database of indigenous drugs. Existing resources are scattered across textbooks, not optimised for mobile access.



No Community Contribution Pipeline

Local communities hold valuable regional knowledge that is never captured. There is no peer-validated mechanism for community members to contribute and verify new drug entries.

Project Objectives

Four core goals driving the application



AI Leaf Identification

Enable users to photograph a leaf and instantly identify the medicinal plant using an on-device TFLite CNN model — no internet required for inference.



Curated Drug Database

Provide a searchable, Firebase-backed database of 31+ indigenous plant species with scientific names, medicinal uses, and preparation methods.



Community Submission & Validation

Allow community members to submit new drug entries, which undergo an admin peer-review pipeline before being published to the live database.



Multi-Language Accessibility

Serve North-East Indian users in English and Assamese (অসমীয়া) with dynamic runtime locale switching via LanguageManager.

Technology Stack

Tools, languages, and frameworks used



Android Studio

Hedgehog / Iguana IDE



Java 8

Source language



Kotlin DSL

Gradle build scripts



TensorFlow Lite

v2.16.1 — On-device AI



Firebase Auth

Email + Google OAuth 2.0



Firebase Realtime DB

NoSQL JSON tree



Firebase Storage

Images & media files



View Binding

Type-safe view access

System Architecture

Three-tier design: Public → Prediction → Admin

TIER 1 — Public Layer

- Plant / Drug Database browsing
- Search & filter medicines
- Multilingual UI (EN / Assamese)

TIER 2 — AI Prediction Layer

- Leaf photo → TFLite CNN
- 31 species, on-device inference
- < 500 ms response time

TIER 3 — Admin / Backend

- Drug submission validation
- User management
- Subscription control

Firebase Backend



Firebase Authentication



Realtime Database (JSON)



Cloud Storage (media)



Firebase Analytics



AI / ML Module

Leaf Prediction Engine

Framework: TensorFlow Lite 2.16.1

Input: 150 × 150 px RGB

Output: Softmax / 31 classes

Threads: 4 CPU (parallel)

Inference: < 500 ms

Accuracy: 85-95% (top-1)

Inference Pipeline

1 Capture / Select Leaf Image

2 Scale Bitmap to 150 × 150 px

3 Normalize pixels ÷ 255 → [0.0, 1.0]

4 ByteBuffer → TFLite Interpreter (4 threads)

5 Softmax output → ArgMax → Top class

6 Display: Plant Name + Confidence %

Firebase Backend (BaaS)

No custom server — all managed by Google Firebase



Firebase Authentication

- Email & Password login
- Google Sign-In (OAuth 2.0)
- Auto user profile creation
- Session persistence



Realtime Database

- Cloud-hosted JSON tree
- drugs/ — master records
- drug_to_be_validated/ — pending
- Peoples/ — user profiles



Cloud Storage

- Leaf prediction images
- Drug / plant photos
- Profile pictures
- URLs stored in DB records



Drug Validation Pipeline

- User submits → pending node
- Admin reviews in-app
- Approved → moved to drugs/
- Rejected → entry deleted

Key Features & App Screens

19 activities across 11 core screens



AI Leaf Scanner

Snap a leaf — get plant name + confidence % in < 500 ms. On-device CNN, works offline without internet.



Medicine Library

RecyclerView of all plants with Glide-cached images, scientific names, and medicinal uses from Firebase.



Community Submission

Submit new drug entries via a form. Admin validates the entry before it is published to the live database.



Bilingual Interface

Toggle between English and Assamese (অসমীয়া) at runtime using LanguageManager locale switching.



Subscription & Payment

Three tiers — Free, Basic, Premium — with a mock card payment gateway and Firebase subscription update.



Admin Panel

Admin accounts access UsersActivity to view all registered users, subscription status, and drug validation queue.

Results & Project Metrics

At a glance — what was delivered



19

Activities
(Screens)



43

Java Source
Files



31

Plant Species
(AI Model)



2

Languages
Supported



85–95%

AI Top-1
Accuracy



3

Firebase
Services

All Deliverables — Completed

- ✓ Firebase Auth (Email + Google OAuth)
- ✓ Realtime Database + Cloud Storage
- ✓ TFLite Leaf Classifier (31 species)
- ✓ Drug submission & admin validation
- ✓ Subscription tiers + mock payment
- ✓ Multi-language (English + Assamese)

APK Build Info

Build Type:	Debug
Size:	~311 MB (debug)
Min Android:	API 27 (Android 8.1)
Target Android:	API 34 (Android 14)
Build Date:	April 23, 2026

Future Work & Enhancements

Roadmap for the next phase of development

01 Training Accuracy Report

Export precision, recall, F1-score and confusion matrix from the Python training notebook for all 31 classes.

02 Release APK Build

Enable ProGuard / R8 code shrinking to reduce the APK from ~311 MB to an estimated 40–60 MB.

03 Expand Plant Database

Scale the TFLite model to 50+ species with training data from North-East Indian indigenous flora.

04 Offline Caching (Room DB)

Cache Firebase drug records locally using Room so the app remains usable without internet.

05 Firebase Security Rules

Harden database access rules to restrict unauthorized reads/writes and protect community data.

06 AI Feedback Loop

Allow users to flag incorrect predictions, building a feedback dataset to retrain and improve accuracy.

Conclusion

Arogya Mitra

demonstrates how modern mobile technology can preserve and democratise indigenous medicinal plant knowledge for the next generation.

- ✓ On-device AI leaf identification (31 species, 85–95% accuracy)
- ✓ Firebase-powered community drug database with validation pipeline
- ✓ Multi-language support (English + Assamese) for North-East India
- ✓ Complete subscription & admin system — 19 screens, 43 source files